

Titanic Exploratory Data Analysis (EDA) Report

Data Analyst Internship — Task 5

Project	Exploratory Data Analysis on Titanic Dataset
Tools Used	Python • Pandas • Matplotlib • Seaborn
Dataset	Titanic (891 records, 12 features)
Report Type	Internship Submission — Task 5

1. Introduction

The purpose of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset and identify patterns, relationships, and trends that influenced passenger survival. The analysis was conducted using Python, Pandas, Matplotlib, and Seaborn.

2. Dataset Overview

Metric	Value
Total Records	891
Total Features	12
Numerical Features	7
Categorical Features	5
Missing — Age	177
Missing — Cabin	687
Missing — Embarked	2

3. Visual Analysis and Findings

3.1 Survival Distribution

Observation:

- More passengers died than survived.
- The number of non-survivors exceeded the number of survivors significantly.

```
plt.figure(figsize=(6,4))
sns.countplot(x='Survived', data=df)
plt.title("Survival Distribution")
plt.show()
```

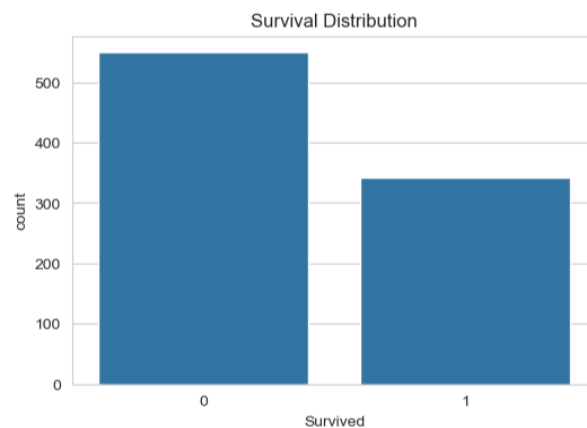


Figure: Survival Distribution

3.2 Gender vs Survival

Observation:

- Female passengers had a considerably higher survival rate than male passengers.
- Most male passengers did not survive.

```
plt.figure(figsize=(6,4))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title("Gender vs Survival")
plt.show()
```

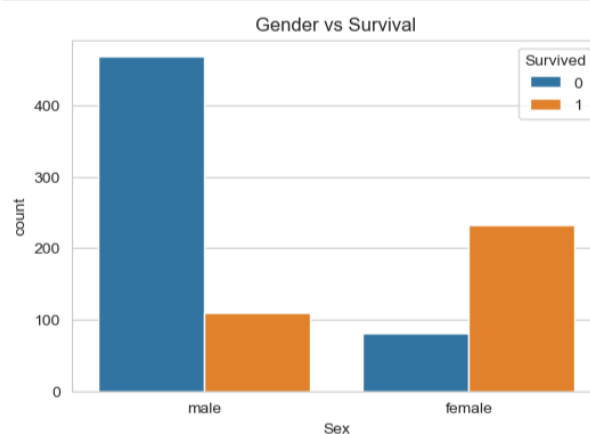


Figure: Gender vs Survival

3.3 Age Distribution

Observation:

- Most passengers were between 20 and 40 years old.
- The age distribution is slightly right-skewed.

```
plt.figure(figsize=(8,4))
sns.histplot(df['Age'], bins=30, kde=True)
plt.title("Age Distribution")
plt.show()
```

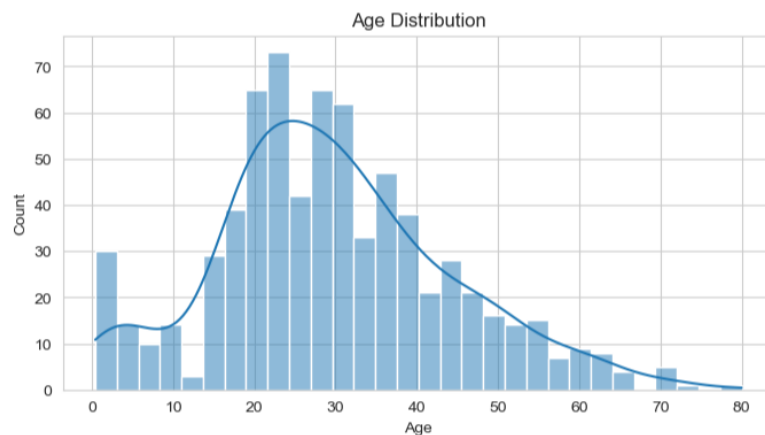


Figure: Age Distribution

3.4 Fare Distribution

Observation:

- Fare values contain multiple outliers.
- Some passengers paid exceptionally high ticket fares.

```
plt.figure(figsize=(8,4))
sns.boxplot(x=df['Fare'])
plt.title("Fare Distribution")
plt.show()
```

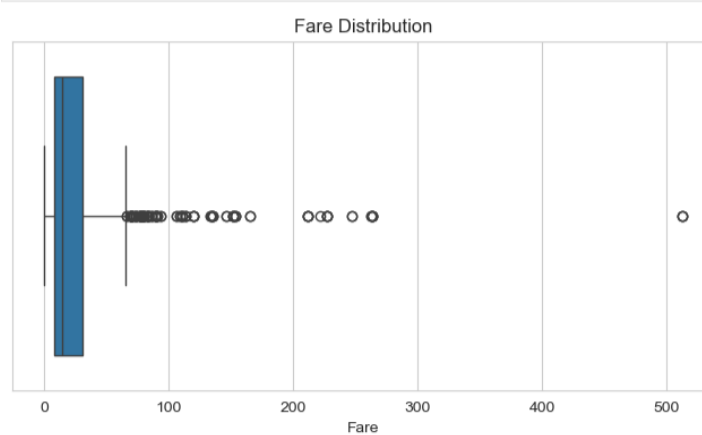


Figure: Fare Distribution

3.5 Passenger Class vs Survival

Observation:

- First-class passengers had the highest survival rate.
- Third-class passengers experienced the highest number of deaths.

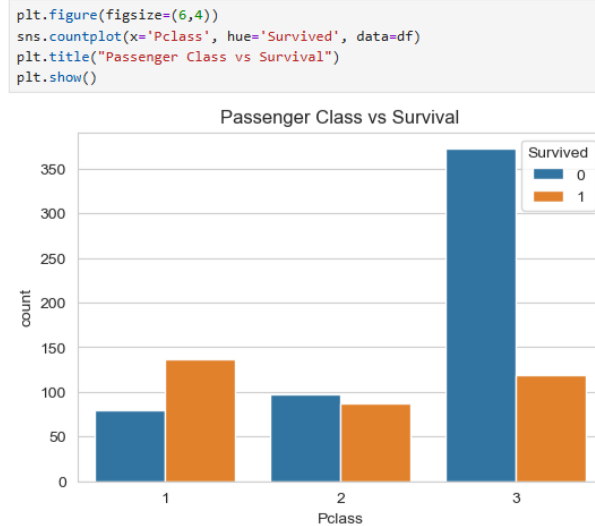


Figure: Passenger Class vs Survival

3.6 Correlation Analysis

Observation:

- Fare and survival show a positive relationship.
- Passenger class and survival show a negative relationship.
- Fare and passenger class have a strong negative correlation.

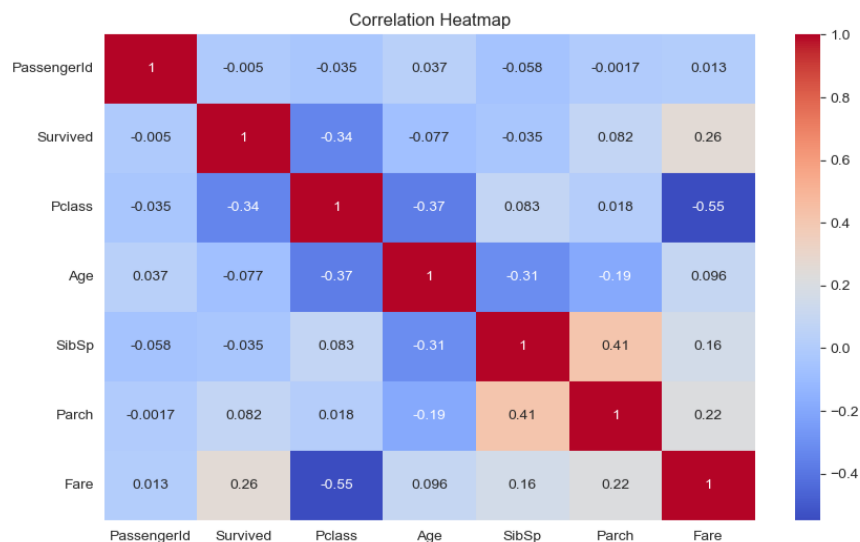


Figure: Correlation Heatmap

3.7 Pair Plot Analysis

Observation:

- Fare values are concentrated at lower ranges with several extreme values.
- Passenger class influences fare and survival patterns.
- No strong linear relationship exists between age and survival.

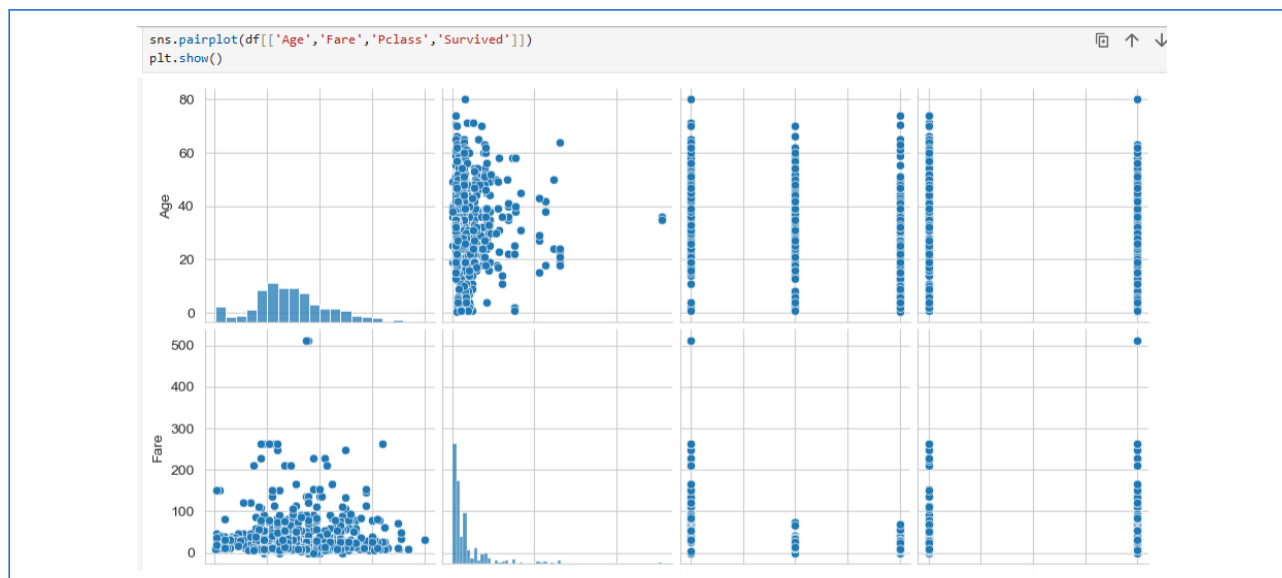


Figure: Pair Plot

4. Key Findings

- Female passengers were more likely to survive than male passengers.
- First-class passengers had better survival chances than lower-class passengers.
- Most passengers were between the ages of 20 and 40.
- Fare data contained significant outliers.
- Missing values were present in Age, Cabin, and Embarked columns.
- Passenger class and fare were important factors affecting survival.

5. Conclusion

The Exploratory Data Analysis of the Titanic dataset successfully identified important trends and relationships influencing passenger survival. Gender, passenger class, and fare emerged as significant factors associated with survival outcomes. The analysis also revealed missing values and outliers that should be addressed during future predictive modeling tasks. Overall, EDA provided valuable insights into the dataset and demonstrated the importance of data exploration before advanced analysis and machine learning.